# HIPE Astronomer Acceptance Test Report

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ABSTRACT

For seven months I have worked as an HIPE Astronomer Acceptance Tester at the Herschel Science Center. During this period, previous work on the Chamaeleon and Musca molecular clouds complex was continued inside the context of the Gould's Belt Key Program (KPGT_pandre_1). Different map making algorithms were tested on these regions, which provided useful information on the performance of these algorithms when working with complex maps like these ones. Some prelimiary results on photometric comparisons between the algorithms were also obtained, so far pointing to a good agreement among them. Up to this date, 24 SxR have been raised, some of them leading to substantial changes on different aspects of HIPE. I also participated part in AAT8 and AAT9 as part of my AAT contract, helping the respective HIPE versions to be as good as possible.

As a short summary, I found:

- HIPE has improved a lot from version 6 to its current state. The pipelines behaviour, memory consumption and results are much better now, and the documentation is much more extense and clear.

- However, there are still some details to improve: special efforts should be made to lower the memory requirement, improve the photometric measurements and provide reliable error maps for Herschel data.

- HIPE will be a very useful tool for me in the future, not only to deal with Herschel data, but also to treat fits files and use python with an interactive programming environment.

# 1 Introduction

For the last seven months, I have worked as a HIPE Astronomer Acceptance Tester at the Herschel Science Centre at ESAC. This fact, combined with the expertise I acquired during my traineeship, has given me the opportunity to get an idea of how a "first time" user feels when working with HIPE, and to find some of the *pros and cons* of the software as an astronomer. Besides, I have been able to test the evolution of HIPE from March 2011 (version 6) to the present (version 9), which also gave me an impression of how the software is developing.

In my case, I worked with observations from the Chamaeleon and Musca complex obtained in parallel mode (PACS & SPIRE observations, with durations between 2 and 4 hours each map, double time when combining scan and cross-scan with total areas of ∼3-7 square degrees). Since these maps are considerably big, they require an important amount of memory for being reprocessed (a 16Gb computer would crash for the PACS blue channel, and PACS data took some hours to be processed), so I also had to deal with the *batch mode*, the configuration of HIPE within the context of a cluster of computers (scigrid) and not being able to use the graphical user interface. I focused on the processing of these regions with different algorithms for the Gould's Belt Key Program (KPGT_pande_1). Therefore, I have some ideas about what is working and what is not for different map making algorithms.

And as the last part of my testing, I started with some photometric comparisons for the different maps. Although this study is still in a very early stage, I have found some problems within HIPE that I consider fundamental and could result in some important improvements for the future. I have also participated actively in the Astronomer Acceptance Testing phase of HIPE 8 and HIPE 9 in the following areas of the framework testing: HIPE GUI, IMAGE and PLOTTING.

In this report I intend to summarise these seven months as a HIPE Astronomer Acceptance Tester. I will describe some of the most important parts of my experience with HIPE, discuss the preliminary results of the photometry study and provide a list of the SxRs opened during this period.

# 2 HIPE as an astronomer

I started using HIPE when the operational version was the 6th one. At that point I had just finished my degree, so not only was this my first experience with HIPE, but also in astrophysical research. There is no need to say that from that point to the actual 9th version, things have change substantially. I had my first contact with the software at the Herschel Open Time Cycle-1 Data Processing Workshop (14-18 March 2011), and from that point on I have been working at ESAC, constantly surrounded by the experts of the Herschel Science Centre. In this sense, I guess my experience can not be considered as a standard one, since I have always had somebody around to help with my problems. However, given the fact that I had no previous technical knowledge of the instruments nor the software, I think most of the astronomers would have experienced the same problems I had.

As I said, things have changed a lot since I started using HIPE, and in my opinion they have done it in the correct way:

- Not only has the software become much more efficient and clear in the sense of "understandable" for the user, but the documentation as well. I find this is one of the most important improvement that could have been done: documentation is vital for somebody working outside "the HIPE environment". It is the first place (and most of the times, the only place) where someone will search for help. Considering the state of the documentation and comparing it with the documentation available when I started, I can totally say it is tremendously easier today to find what you need. It is much better structured, there is more information and the most important thing: there are examples for all the usual things one would like to do within HIPE. For me there is no doubt regarding this: the better the documentation is, the easier HIPE will be. Therefore, I am also sure that the process of learning HIPE today is much more painless than it was a year ago.

- The memory consumption is decreasing. Memory has always been one of the most important problems of HIPE, to the point of preventing some observations to be processed unless one had access to considerabily powerful computers. Even if HIPE did not crash, algorithms like photProject or madMap used to take a lot of time or freeze if the total amount of RAM memory was used and the SWAP one was required. The algorithms have improved, reaching acceptable levels of memory and time requierements.

- Following the same tendency, algorithms have improved their performance. As an example, situations like poor baseline removal induced overshooting regions that

alter the maps, making them simply unusable. The development of algorithms like madMap for PACS or the implementation of destriper in SPIRE have meant an extremely important progress on the processing of complicated regions like the ones considered here, solving these problems.

However, it is more important to focus on what should still be changed. It is to be expected that a software of this kind has several "small details" that could be modified, but they do not affect the overall functionality of the program. I find HIPE is close to this stage, but has not reached it yet. In my opinion, there are a few points of vital importance that I think should be corrected if the goal is to turn HIPE into something that people use not only for the processing but for analysis as well. To say more, even if HIPE is going to be used only for processing, there are some aspects that need improvement so that users consider HIPE a friendly software. As I see it, these points really demand more effort:

- First of all, I think the memory usage of HIPE (specially of PACS) is something that should be critically considered. I am not an expert on the pipelines or the instruments, and I do understand that behind them there are extremely complicated processes taking place. Even with this situation, the memory requirement to process PACS maps directly implies that some of the observations will not be processed just because of lack of resources. This is not a problem for small maps or mostly clear regions (where the archive output is good enough), but when it comes to filamentary structures and dusty clouds things are different: these data require special handling. If more than 30 Gb are required, then most of the users will give up (in the case of the Chamaeleon and Musca complex, half of the PACS observations requiered these amounts of memory to be processed). I do not know if there is a way to make the processing more efficient in a "mathematical" way, but I am sure there are small details that could be improved and will improve the performance of HIPE. As an example, something as simple as deleting those variables that are not required anymore decreases the memory usage of the madMap algorithm substantially. SPIRE handles more tolerable levels of memory (although they can be challenging for big maps as well), but the case of PACS is, in my opinion, quite serious. I would consider this as a high priority task for the next HIPE version.

  In the case of this problem being extremely complicated to solve, there could be partial solutions that may make this issue more bearable:

  - I would include a small routine at the beginning of each pipeline that would rapidly go through the pipeline and give an estimation of how much memory/time would be required to process the observation with the current

script. Another version of the previous suggestion could be to modify HIPE to compute the memory required for the next step before running it. This will guarantee that HIPE does not freeze while running a script.

  - I would make a detailed analysis of the latest versions of the pipelines and add an additional option to all of them: if the user thinks memory could be a problem, the pipeline would delete every variable once the script reaches the point where that variable is not needed anymore (SCR: PACS-4206).

- Another important problem of the nowadays status of HIPE is the photometry. I have met several people starting with HIPE and the most common question was "How do you extract photometry from Herschel maps?". However, there is no answer to that question yet. There are different source extraction algorithms implemented in HIPE like *Daophot* and *Sussex*, but none of them works in such a way that we could guarantee they will work for every map. There are alternative problems, like the output of these algorithms being not interactive at all (once the sources are extracted, it is probably necessary to select them, delete duplicates,...), or all the artificial sources detected at the edges of the images (and since there is no way to interact with the source list within HIPE, there is no way to remove them). Another example could be the performance of these algorithms with madMap maps, where instead of "NaNs", zero values are used: as a result, the statistics on the image (and the source extraction process) become a mess, making the output unusable. This is another important point to modify: HIPE should offer a reliable method to extract the sources. Otherwise, even if the user has a powerful enough computer to process the data, there will be no way to use them efficiently (the user sill simply not extract a catalogue from a big map source by source) (SCR: HCSS-15923).

- In a tight relationship with the previous point, the error estimation problem is another big deal. As an astronomer, I depend completely on errors to verify and quantify the goodness of my analysis. Therefore, users won't trust Herschel results if they can not estimate these errors. The error maps from PACS are simply wrong, and the methods used to make an estimation (like calculating the rms in the nearby zones) are quite artificial and not accessible for everyone (it can be pretty hard for an external user to create a script to do this). To say more, the errors provided by the simplest photometric task implemented in HIPE (aperture photometry) are prepared for CCDs, so they are unusable as well. I once got into a discussion with some of the PACS experts on this topic and it certainly is a complicated problem. Even with this case, I feel it would be worth making all the effort required to solve the it.

- There is an additional point which, although more difficult to solve, could also be

important to treat: the number of map making algorithms and parameters on the scripts. Given the different "types" of regions one can find (clean isolated sources, star forming regions, filaments, bubbles,...) it is to be expected to find different ways to process them. However, this is extremely confusing for people who are not used to HIPE. The most relevant case is found for PACS and the difference between photProject and madMap. I learnt what each of them is for, but from my perspective, it makes no scientific sense to have two different methods to process the maps: as an astronomer I want my data to be processed in the correct way, and it is shocking to have two ways that give different results and accept both of them are correct (sometimes, the results are substantially different). Even "inside" a particular algorithm, there are some parameters (high-pass filter,nsigma,...) which alter the output as well. I am aware it is not possible to find an absolute way to process all the observations (although that would solve several problems for the users), but aiming for the most standard algorithm possible would result, without a doubt, in an improvement in the HIPE experience a new user would have. I think this issue is the main reason why people would use external software like *Scanamorphos* instead of HIPE directly: the output depends on a small number of parameters and most of the times there is no need to use them. This fact makes it simpler and the results are quite good as well, so the users find no benefit in using HIPE.

# 3 Chamaeleon an Musca: dealing with complex maps.

Since I started my traineeship I have been working with the Chamaeleon and Musca molecular clouds complex. This is a set of four molecular clouds, each of them with different star formation rates and physical scenarios. Regardless of their different properties and morphology, the four of them have something in common: complex dusty structures like filaments and bubbles are present all over the regions. These structures tend to produce regions with strong overshooting due to the baseline removal process. The simple (but challenging) result of this situation is that these maps require a more complex data reduction process to avoid this problem. An example of one of this regions an their complexity can be found in the following figure.

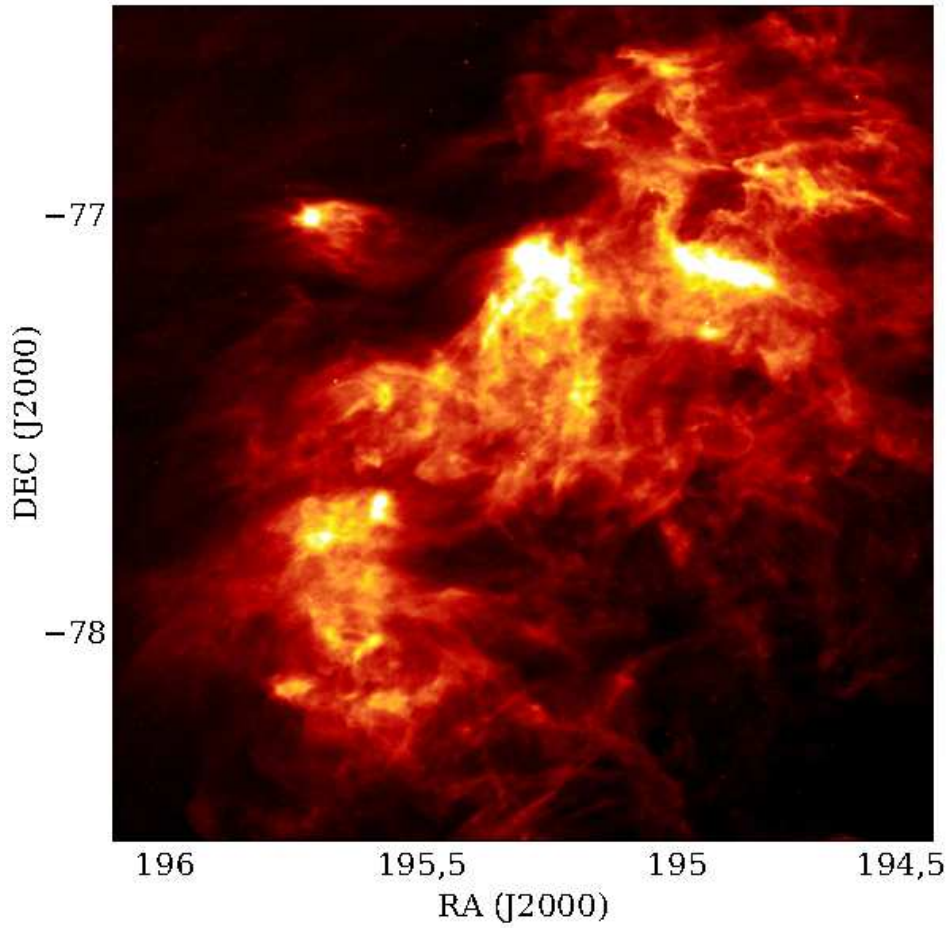



Figure 1: Cha II as seen by SPIRE (250 $\mu$m) processed with Scanamorphos. The intrincate filametary structures can be appreciated in this image. This whole observation required 4 hours of observind time and covers $\sim$3.5 deg$^2$.

## 3.1 PACS

In my experience, PACS data are the most difficult ones to process when it comes to Herschel maps. From the beginning I dealt with considerably big regions, which meant memory problems. Besides, as commented before, these regions show filamentary structures which are really interesting to study, but not trivial to process. In the case of PACS, the pipelines are strongly dependent on some parameters, which makes the processing a little hard for maps with structures.

When I started processing these maps, only photProject was ready. For this specific algorithm, the value of the high-pass filter is essential for defining how the extended emission would be treated or removed. It turned out that photProject maps were not suitable for studying the structures in the maps, but only the clean point sources (there were problems even for those point sources embedded in a small cloud, since it is not possible to know how much of the removed extended emission actually belongs to the source).

madMap was not ready at that time, but it has evolved to be the best choice (within HIPE) for extended emission. Some troubles were found at the beginning since the "nsigma" parameter, responsible of the behaviour of the deglitching process, appeared to have a excessive impact on the maps. However, this problem has already been solved. Besides, its treatment of the 1/f noise left a cleaner background than photProject, and the extended emissions were recovered much better. However, I would like to point out that even though the output maps looked great, it did not guarantee they were properly processed. With help from Babar Ali, we took a look to the behaviour of the temperature drift value for different parts of the detector, finding that all of them should be processed in different ways, depending on the behaviour of these drifts (two examples of this problem can be found in the next page). It is therefore very important to warn the users about these kind of problems, and all the sanity checks that should be done before using madMap, or otherwise it is quire probable that "wrong" maps are used for the scientific analysis.

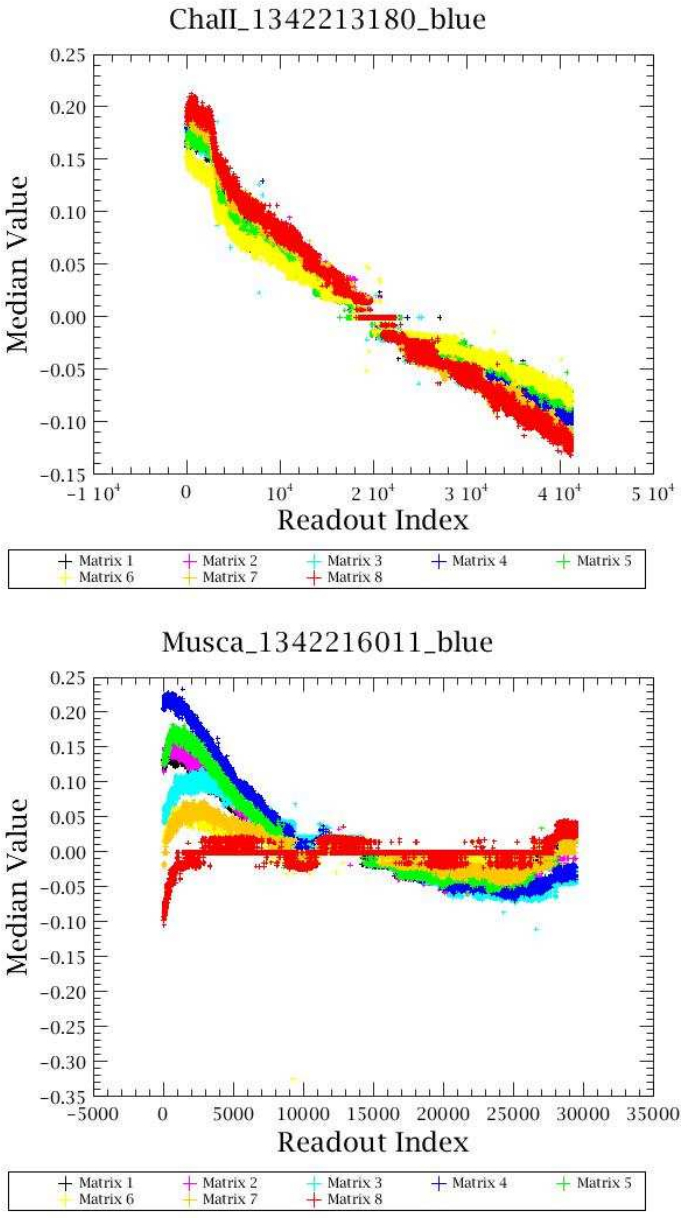



Figure 2: Module to module drifts for some of the regions. As it can be seen, there is a jump at the start of the frames for ChaII, and the first ∼2000 readouts of Musca should be ignored in order to get a proper temperature drift correction. madMap expects flat or smooth curves in order to work properly. This simple diagnostic should be explained to users reducing PACS maps with madMap.

## 3.2 SPIRE

The case for SPIRE maps was somehow similar when I started processing these regions: there was one algorithm ready (naiveMap), and another one was being developed (destriper).

naiveMad and madMap are exactly the same algorithm for SPIRE maps, since they do not suffer from 1/f noise. The resulting maps in the case of the Chamaeleon and Musca complex did not look completely good. The existence of the filamentary structures made the baseline removal process much more complicated, and the maps suffered from overshooting regions.

The solution came with destriper. It is not exactly an algorithm itself, but an additional way to remove the baseline in a much better way. The background was cleaned from the overshooting regions and the maps were ready to do science with them.

## 3.3 Scanamorphos

Although Scanamorphos is an external software I think it is worth saying some words about it. Apart from the previously commented algorithms, Scanamorphos was also tested both for PACS and SPIRE maps and it is fair to say that it was found to be the best one so far, not only in terms of time and memory consumption (specially for PACS: only for the blue channel a 32 Gb computer was required to process two of these regions) but also when the quality of the maps is considered. It was capable of dealing with the problem of the drifts for PACS without any special treatment, and produced maps at least as good as destriper for SPIRE. This, combined with the fact that there are almost no parameters on which the behaviour of the algorithm depends would probably make it the first choice for me. I definitely think that even with the respective problems, an effort should be made to turn the scripts within HIPE into the most straight forward algorithms as possible.

## 3.4 Photometry

Currently, it is still not clear for a user what the best procedure to extract photometry from Herschel maps is. I have had several problems trying to figure out what the best way to get it is, since every source extraction algorithm treats the maps in a different way. A quick visual inspection reveals that both *Daophot* and *Sussex* find most of the clear sources (there is no easy way to quantify this). However, they also detect a very high amount of sources on the filamentary structure. These are likely not to be real (a very large amount of them could in fact be protostars, and it is not a trivial problem to separate between the two cases). For a high-level photometric procedure, more complex algorithms external to HIPE like getSources (developed by the Gould's Belt Key Program team) can be used in order to use information from all the bands, detect filaments themselves,... In my case, I just intended to check differences between photometric values in different maps, so there is no need of such a complex algorithms: I simply selected those sources at (more or less) clean regions, and compared them in both maps.

I found some trouble with madMap maps, probably due to the zero values used outside the map to build the matrix: the whole map got completely covered when any of the two algorithms were applied. The zero values at the edges are likely to alter the statistics on the image, producing this result (SCR: PACS-4451). On the other hand, Sussex is unable to deal with madMap maps: it requires an error maps, and the ones given by madMap only have zero values (SPR: PACS-4472). Therefore, Daophot is used for the comparison here. In order to check the robustness of the comparison, this procedure is done in two smallest regions of the complex: ChaII and Musca. The red channel of PACS and the PSW camera of SPIRE were used. Once the "clean" regions were selected (using Topcat) the crossmatch was done using a search radius of 12" and 18" respectively. A detection level of 5 (equivalent to a $\sim 5\sigma$ level) is established.

| **PACS 160** | **ChaII** | **Musca** |
|---|---|---|
| photProject vs Scanamorphos | 3 common sources | 8 common sources |

| **SPIRE 250** | **ChaII** | **Musca** |
|---|---|---|
| naiveMap vs destriper | 0 common sources | 0 common sources |
| naiveMap vs Scanamorphos | 0 common sources | 0 common sources |
| destriper vs Scanamorphos | 27 common sources | 27 common sources |

Before going into the comparison results, I want to point out two facts that I consider important. First of all, the number of sources at the PACS maps can be considered as "low". However, the strict conditions imposed should be considered: the source

must have be $\geq 5\sigma$ detection and it has to be located outside the filaments and clouds. For SPIRE, a whole background of galaxies is found, but not in PACS maps. To say more, the background of PACS maps are usually much noisier than SPIRE ones. The relevant thing here is that these sources are isolated and clearly detected, so it is to be expected that they match quite well. The second point is the impossibility to extract a reliable whole catalogue of the SPIRE naive maps: due to the previously mentioned overshooting regions, the whole source extraction is altered and most of the sources are located on these zones (these sources are, of course, non-real sources but artifacts). As a result, when the crossmatch with sources from the clean parts of the region (from another map, i.e. destriper) was performed, no matches were found. The conclusion of this would probably be that in the presence of filaments and structures that could alter the baseline removal process, naiveMap maps can not be used to obtain a reliable catalogue: destriper is mandatory in these cases.

The total amount of sources are combined to give a more relevant statistics. The results are showed in the following figures:

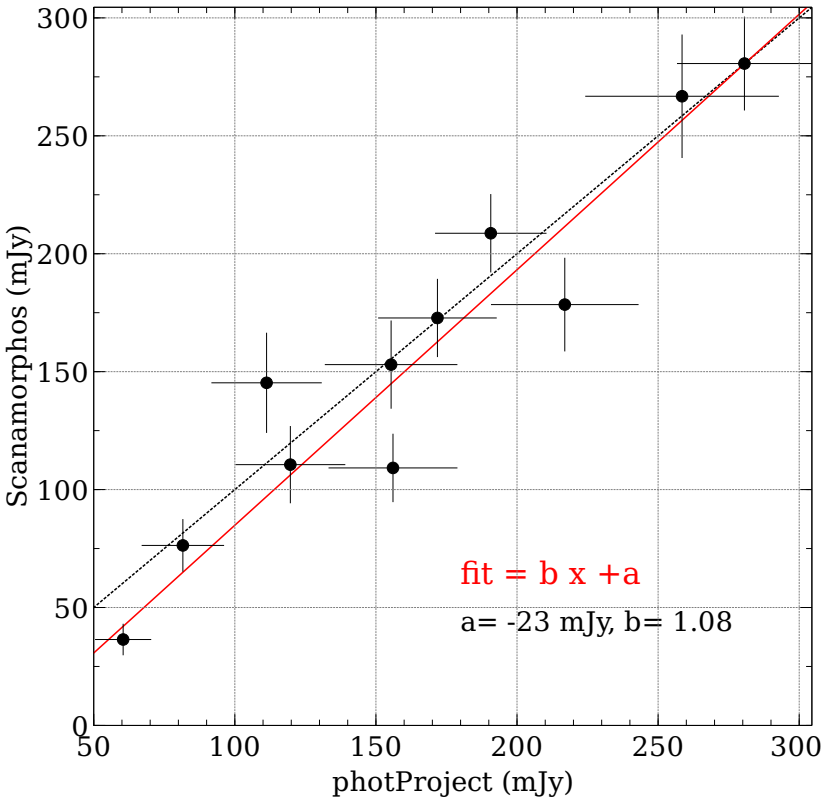



Figure 3: Combined PACS photometry in both maps. The fit is plotted in red. The

y=x line is plotted as a dotted black line for comparison. The errors are calculated directly by the *Daophot* task.

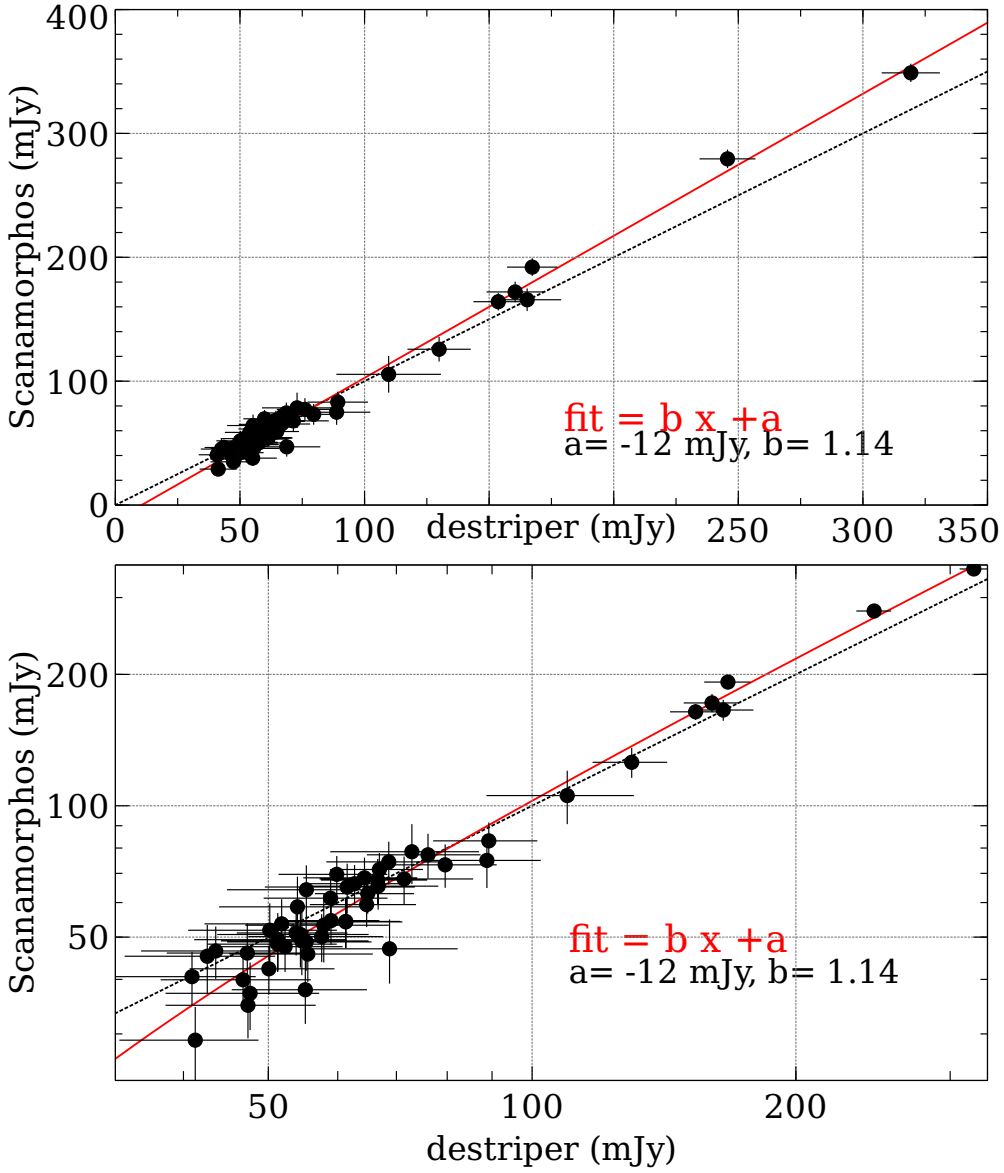



Figure 4: Combined SPIRE photometry. The fit is plotted in red. The y=x line is plotted as a dotted black line for comparison. The lower graph shows the same results in log-log scale to detail the sources with lower flux levels. The errors are calculated directly by the *Daophot* task.

The results are as follow:

- For PACS, the fluxes are comparable within a 8%. Scanamorphos seems to give slightly lower values than photProject. However, we should not forget the effect of the high-pass filter applied on the photProject map and the modifications it can cause on the flux measurements. It is important to notice that the errors provided here (and used for the fit) are the ones given by the algorithm, which does not calculate actual errors. Anyway, the correspondence of the fluxes is well inside the estimated real 15-20% errors in the fluxes.

- For SPIRE, the results are similar. Once again, the fitting depends on the errors, which are not reliable yet. The fluxes are comparable within a 15%, which although is almost two times more than in the case of PACS, it is still acceptable if compared with the expected errors. The differences could be either a real effect or an induced one: although the selected regions are mainly free of filaments and clouds, it is still possible that some sources are not totally isolated or real ones. Even with this issue altering the statistics, the fluxes are compatible.

- For both of them, the offset is considerably small ($< 25\,\mathrm{mJy}$).

- Most of the sources considered here in these regions are"weak", and none of them has a flux over 1 Jy. These results are likely to improve for brighter sources, where the effect of the background and the noise is lower.

There are some other comparisons that could be made in order to check for the consistency of fluxes within different maps. For example, one could use photometry of single sources to compare the data by hand. This would also solve the problems with madMap maps and the catalogues from naiveMap. However, if a real comparison is to be made, errors are of vital importance to establish *how similar* the outputs from different maps are. However, even in the early stage of the analysis performed here, the results suggest that the maps created using different map making algorithms are compatible within the errors for point source photometry.

**Acknowledgements**

I would like to truly thank Leo Metcalfe, Bruno Merín and the Herschel Science Center for giving me the opportunity and financial support to continue working within the Herschel context.

| List of raised SxR | | | | |
|---|---|---|---|---|
| **Key** | **Issue Type** | **Title** | **Status** | **Resolution** |
| HCSS-15923 | SCR | Interactive task to visualize / modify a source list | Issued | UNRESOLVED |
| HCSS-15909 | SCR | DPAT9: Annular and RectangularSkyAperturePhotometry tasks not working in the "All" folder | Issued | UNRESOLVED |
| HCSS-15908 | SCR | DPAT9: Daophot source extraction algorithm task should not ask for the beamArea as mandatory | Closed | Withdrawn |
| HCSS-15907 | SCR | DPAT9: Error in an line of code in the Data Analysis Guide | Issued | UNRESOLVED |
| HCSS-15906 | SCR | DPAT9: Example not working on the Data Analysis Guide | Issued | UNRESOLVED |
| PACS-4472 | SPR | DPAT9: Sussextractor problem when running on a madMap map | In Analysis | UNRESOLVED |
| PACS-4462 | SPR | Strongly different behaviour of source extraction algorithms when using a photProject/madMap map | Issued | UNRESOLVED |
| PACS-4451 | SCR | madMap zero values at the edges of the maps | Closed | Withdrawn |
| HCSS-15619 | SPR | Scanamorphos plug-in not loading image units | Closed | Fixed |
| PACS-4363 | SPR | HIPE crashes when running in batch mode due to PACS calibration updater | In Analysis | UNRESOLVED |
| HCSS-15409 | SCR | Error warning for not completed observations | Issued | UNRESOLVED |
| SPIRE-4150 | SPR | Negative values with destriper | Closed | Withdrawn |
| PACS-4206 | SCR | madMap memory usage | Closed | Fixed |
| HCSS-15087 | SPR | Missing pipeline when installing HIPE 8.0.1 from the CIB installer | Closed | Fixed |
| HCSS-15080 | SPR | Mosaic task not orienting the images | In Analysis | UNRESOLVED |
| HCSS-15078 | SPR | Problem when building a mosaic with big regions | Issued | UNRESOLVED |
| HCSS-15051 | SCR | False source detections at the borders of parallel mode observations | Issued | UNRESOLVED |
| HCSS-14981 | SCR | More documentation needed for the multiplot documentation | Issued | UNRESOLVED |
| HCSS-14796 | SCR | DPAT8: Documentation of the histogram plotting section incomplete | Issued | UNRESOLVED |

Continues in the following page.

| List of raised SxR | | | | |
|---|---|---|---|---|
| **Key** | **Issue Type** | **Title** | **Status** | **Resolution** |
| HCSS-14773 | SCR | DPAT8: ImageCeil, imageFloor and imageRound tasks do not allow the user to decide which decimal is to be rounded. | Issued | UNRESOLVED |
| HCSS-14771 | SCR | DPAT8: Incomplete sentence in the documentation | Closed | Fixed |
| HCSS-14770 | SPR | DPAT8: Transpose task asks for an exposure plane | Validating | UNRESOLVED |
| HCSS-14769 | SPR | DPAT8: The annularSkyAperturePhotometry tasks does not allow to move the aperture | Analyzed | UNRESOLVED |
| SPIRE-3884 | SPR | DPAT8: Crash of sourceExtractorSussextractor task | Closed | Withdrawn |